

SARAH SKINNER

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Education

Carnegie Mellon University (CMU)

PITTSBURGH, PA

Doctorate in Physics, 3.9/4.3

Aug 2024

Missouri University of Science and Technology (Missouri S&T)

ROLLA, MO

Bachelor's Degree in Physics, *Minors in Computer Science, Mathematics*, 3.9/4.0

Dec 2019

Skills

Programming Languages: C++, PYTHON, FORTRAN, LATEX, BASH/BATCH, SQL, HTML5/CSS

General Skills: PARTICLE PHYSICS, STATISTICAL ANALYSIS, ACADEMIC WRITING, PROFESSIONAL MENTORING

Applications/OS: MICROSOFT OFFICE, MATHEMATICA, LINUX, WINDOWS

Primary Programming Methods: MPI, OPENMP, MONTE-CARLO SIMULATIONS, SCIPY, NUMPY

Projects

Ansys Mechanical Linux AArch64 Port (Synopsys, Inc.)

CANONSBURG, PA

Supervisor: *Jeff Beisheim*. Facilitated and supported porting the Ansys Mechanical solver to Linux AArch64 systems.

Aug 2024 - Present

Calculating Resonance Information from Lattice QCD (CMU) 

PITTSBURGH, PA

Advisor: *Dr. Colin Morningstar*. Utilize statistical analysis to study simulation data to compute scattering observables to understand how resonances affect scattering results.

Jan 2021 - Aug 2024

Work with large collaborations to coordinate efforts and publish papers.

Scalability Improvement of Plasma Simulation (NASA Glenn Research Center)

CLEVELAND, OH

Advisor: *Dr. Maria Choi*. Improved parallelization methods and optimized code within a hybrid fluid/Monte Carlo plasma simulation in confined geometries to predict behavior around an ion thruster.

Jan 2020 - May 2020

Anomalous Diffusion in Confined Geometries (Missouri S&T) 

ROLLA, MO

Advisor: *Dr. Thomas Vojta*. Modeled anomalous diffusion governed by the fractional Langevin equation in confined geometries using Monte Carlo methods in order to understand how serotonin fibers behave in the brain.

Jan 2018 - Dec 2019

Errors in Dark Matter Halos Fit Model (Brookhaven National Laboratory) 

UPTON, NY

Advisor: *Dr. Anže Slosar*. Inserted error into dark matter fit model parameters to assess its significance and determine validity of the original measurements.

Jun 2019 - Aug 2019

Binary Black Hole Model (Louisiana State University) 

BATON ROUGE, LA

Advisor: *Dr. Peter Diener*. Added rotation parameter for smaller black hole in binary system with a high mass ratio to study electromagnetic and gravitational waves from a more realistic black hole system.

May 2018 - Jul 2018

Publications

- [1] **Baryon Scattering (BaSc) Collaboration** Collaboration, J. Bulava, B. Cid-Mora, A. D. Hanlon, B. Hörz, D. Mohler, C. Morningstar, J. Moscoso, A. Nicholson, F. Romero-López, **S. Skinner**, and A. Walker-Loud, *Two-Pole Nature of the $\Lambda(1405)$ resonance from Lattice QCD*, *Phys. Rev. Lett.* **132** (Jan, 2024) 051901.
- [2] **Baryon Scattering (BaSc) Collaboration** Collaboration, J. Bulava, B. Cid-Mora, A. D. Hanlon, B. Hörz, D. Mohler, C. Morningstar, J. Moscoso, A. Nicholson, F. Romero-López, **S. Skinner**, and A. Walker-Loud, *Lattice QCD study of $\pi - \bar{K}N$ scattering and the $\Lambda(1405)$ resonance*, *Phys. Rev. D* **109** (Jan, 2024) 014511.
- [3] J. Bulava, A. D. Hanlon, B. Hörz, C. Morningstar, A. Nicholson, F. Romero-López, **S. Skinner**, P. Vranas, and A. Walker-Loud, *Elastic nucleon-pion scattering at $m = 200$ MeV from lattice QCD*, *Nuclear Physics B* **987** (2023) 116105.
- [4] T. Vojta, S. Halladay, **S. Skinner**, S. Janušonis, T. Guggenberger, and R. Metzler, *Reflected fractional Brownian motion in one and higher dimensions*, *Phys. Rev. E* **102** (Sep, 2020) 032108.
- [5] T. Vojta, **S. Skinner**, and R. Metzler, *Probability density of the fractional Langevin equation with reflecting walls*, *Phys. Rev. E* **100** (Oct, 2019) 042142.

- [6] A. Nicola, D. Alonso, J. Sánchez, A. Slosar, H. Awan, A. Broussard, J. Dunkley, E. Gawiser, Z. Gomes, R. Mandelbaum, H. Miyatake, J. A. Newman, I. Sevilla-Noarbe, **S. Skinner**, and E. L. Wagoner, *Tomographic galaxy clustering with the Subaru Hyper Suprime-Cam first year public data release*, *Journal of Cosmology and Astroparticle Physics* **2020** (mar, 2020) 044.

Conferences and Workshops

Lattice Conference 2023	Fermilab, IL, August 2023
National Nuclear Physics Summer School	Riverside, CA, July 2023
APS Topical Group on Hadronic Physics Meeting	Minneapolis, MI, April 2023
American Physical Society (APS) April Meeting	Minneapolis, MI, April 2023
International HPC Summer School	Athens, Greece, June 2022
Hampton University Graduate Studies Program (Jefferson Lab)	Newport News, VA, June 2022
American Physical Society March Meeting	Boston, MA, Mar 2019
Conference for Undergraduate Women in Physics	Corpus Christi, TX, Jan 2019

Presentations

Lattice Conference 2023	FERMILAB, IL
<i>Talk:</i> Lattice QCD studies of the Δ and $\Lambda(1405)$ baryon resonances and the $K_0^*(700)$ and $a_0(980)$ meson resonances ↗	August 2, 2023
American Physical Society March Meeting	BOSTON, MA
<i>Poster:</i> Fractional Langevin equation with reflecting barrier ↗	March 6, 2019

Invited Talks

Brookhaven National Laboratory (BNL)	UPTON, NY
<i>Seminar:</i> The role of exotic operators in determining the finite-volume spectrum from Lattice QCD and its consequences	August 18, 2023

Awards and Honors

Mellon College of Science Travel Grant	CMU, August 2023
Best Theory Poster ↗	HUGS, June 2022

Community Service

Cat Socialization (Human Animal Rescue of Pittsburgh)	Pittsburgh, PA, 2021 - 2025
St. Patrick Middle School Science Fair Judge	Rolla, MO, May 2019
Science Olympiad Proctor	Missouri S&T, Sept 2018
Science Learning Center Volunteer	BNL, July 2018